A Synopsis

On

WATER QUALITY ANALYSIS FOR IRRIGATION

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In

COMPUTER SCIENCE AND ENGINEERING

By

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PROJECT DETAILS

TITLE OF THE RESEARCH PROPOSAL	WATER QUALITY ANALYSIS FOR IRRIGATION
NAME OF THE STUDENTS/GROUP	Bhoomi Baranwal
NAME AND DESIGNATION OF THE GUIDE	Ms. Varsha Srivastav
PLACE OF WORK	IILM UNIVERSITY, GREATER NOIDA
AREA	Data Analysis
DOMAIN	Water Quality Analysis
CO-DOMAIN	Data Science/ Python
APPLICATION AREA	Agriculture and Irrigation Management
USEFULNESS	To determine if water is suitable for irrigation and to support better crop growth.
ANY OTHER KEY INFORMATION	The analysis is performed using datasets related to irrigation water parameters such as pH, Electrical Conductivity (EC), Salinity, SAR, and dissolved salts.

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- Schematic Flow Diagram of Irrigation Water Quality Assessment.
- Data Cleaning and Processing Steps.
- Analysis of Water Quality Parameters (pH, EC, SAR, TDS).
- Classification of Irrigation Water Suitability.
- Dataset Structure Used for Analysis.
- Gantt Chart of Project Timeline.
- Visualization of pH Values in Irrigation Water Dataset.
- Electrical Conductivity and Salinity Analysis Graph.
- Overall Water Quality Analysis Result Chart.

Statement of the proposal

This project focuses on analyzing water quality parameters to determine whether water is suitable for irrigation. Poor irrigation water quality can negatively impact soil fertility and crop growth. By analyzing important parameters using data analysis tools, the project aims to identify water suitability for agricultural use.

Problem Statement

Poor water quality used for irrigation can lead to soil degradation, reduced crop yield, and long-term agricultural problems. Therefore, it is important to analyze water quality parameters such as pH, salinity, electrical conductivity, and dissolved solids to determine whether water is safe and suitable for irrigation.

Background

Water quality plays an important role in agriculture. Farmers often use groundwater or surface water for irrigation, but the quality of this water varies from region to region. Researchers have studied irrigation water quality using different water quality indices and parameters to assess its suitability for crops. Organizations like the Food and Agriculture Organization (FAO) provide datasets and guidelines related to irrigation water quality which are widely used in research and academic studies.

Motivation / Need for the Research

The main motivation behind this research is:

- To understand how water quality affects crop growth.
- To analyze irrigation water using real datasets.
- To learn data analysis techniques using Python.
- To simplify water quality analysis for academic purposes.
- To help in identifying suitable water for irrigation.

Objective

The main objective of this project is to analyze irrigation water quality parameters and determine whether the water is suitable for agricultural use.

Sub-Objectives

- To study important water quality parameters used in irrigation.
- To collect and prepare a dataset for analysis.
- To clean and preprocess the dataset using Python.
- To analyze water quality and interpret the results.

Mode of Achieving Objective

1. Collect irrigation water quality dataset from reliable sources.
2. Perform data cleaning and preprocessing using Excel and Python.
3. Analyze parameters such as pH, EC, SAR, and TDS.
4. Interpret results to determine irrigation suitability.

Proposed Methodology

The proposed methodology of this project focuses on analyzing irrigation water quality using a dataset and applying data analysis techniques. The study involves collecting water quality data, cleaning the dataset, and analyzing important parameters that affect irrigation suitability.

The main steps followed in this research are:

- Collection of irrigation water quality dataset from reliable sources.
- Data preprocessing and cleaning using Microsoft Excel and Python (Pandas).
- Analysis of important water quality parameters such as pH, Electrical Conductivity (EC), Salinity, SAR, and Total Dissolved Solids (TDS).
- Evaluation of water suitability for irrigation based on standard guidelines.
- Interpretation and presentation of results using tables and charts.

Theoretical Framework

The theoretical framework of this study is based on irrigation water quality assessment principles. The suitability of water for irrigation depends on several chemical parameters that influence soil structure and crop growth.

Important parameters considered in this study include:

- pH: Indicates the acidity or alkalinity of water, which affects nutrient availability to plants.
- Electrical Conductivity (EC): Measures the salinity level of water and determines the concentration of dissolved salts.
- Sodium Adsorption Ratio (SAR): Indicates the sodium hazard in irrigation water and its impact on soil permeability.
- Total Dissolved Solids (TDS): Represents the total amount of dissolved substances in water.
- Salinity: High salinity levels can reduce crop productivity and damage soil quality.

These parameters are analyzed to determine whether the water is safe, moderate, or unsuitable for irrigation.

Sources of Data – Primary or Secondary Data

The project is based on secondary data collected from sources such as *FAO databases and Kaggle datasets.*

Figure 1: Work Flow Diagram

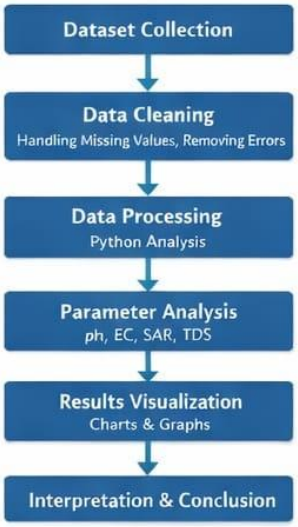


Figure 1: Work Flow Diagram - Water Quality Analysis Process

Figure 2: Schematic Flow Diagram



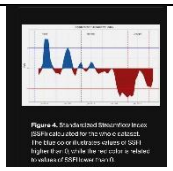
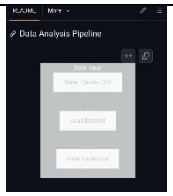
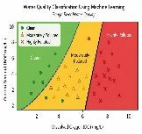
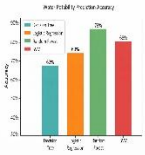
Figure 2: Schematic Flow Diagram - *Irrigation Water Quality Assessment*

Figure 3: Gantt Chart



Figure 3: *Gantt Chart* - *Project Timeline*

Literature Review Table

S.No.	Paper Title (DOI)	Journal/Conference (Year)	1st/Main Author's (Affiliation)	Dataset Used	Code available (Training/ Inference)	Citations	Illustrations	Any Novelty
1.	Data-Driven Water Quality Analysis & Prediction	MDPI (2024)	Silva et al.	UCI Machine Learning Repository – Water Quality Dataset	https://github.com/rfsilva1/data-driven-water-quality	100+		ML model for water quality prediction
2.	Water Quality Testing Data Analysis	GitHub Project	ChanMeng666	Kaggle – Water Quality Dataset	https://github.com/ChanMeng666/water-quality-testing-data-analysis	-		Python data analysis of water parameters
3.	Ganga Water Quality Assessment	GitHub Project	Shivani-781	Central Pollution Control Board (India) Dataset	https://github.com/Shivani-781/Ganga-Water-Quality-Assessment	-		River water quality analysis
4.	Water Quality ML Prediction	GitHub Project	Mohammed840	Kaggle – Water Potability Dataset	https://github.com/mohammed840/Water-Quality-Prediction-machine-learning-python	-		Machine learning classification
5.	Water Quality Monitoring System	GitHub Project	Atharv2808	Sensor Collected Dataset (IoT Sensors)	https://github.com/Atharva2808/Water-Quality-Monitoring-System (Inference)	-		IoT based monitoring system

Research Gaps

Essential Literature Reviewed	Themes drawn from the literatures	Research Gaps	Remark
Data- Driven Water Quality Analysis & Prediction	Use of machine learning models to predict overall water quality using public datasets.	Most studies focus on general water quality prediction, not specifically on irrigation suitability parameters like SAR, EC, and salinity for agriculture.	Scope to develop irrigation- focused analysis model.
Water Quality Testing Data Analysis (Python based)	Data cleaning, visualization, and parameter analysis using Python tools.	Limited integration of long term multi year datasets and lack of region/state comparison.	Opportunity to include multi-scale and multi-dataset analysis.
Water Quality ML Prediction (Kaggle dataset based)	Classification of potable vs non- potable water using government datasets.	Focus is on drinking water quality rather than irrigation standards. Also lacks irrigation indices calculation.	Add irrigation indices such as SAR, EC classification.
Ganga Water Quality Assessment	River water monitoring and pollution analysis using government datasets.	Research mainly targets river pollution, not irrigation usability or agricultural impact.	Can adapt similar analysis for irrigation water sources.
Water Quality Monitoring System (IoT based)	Real-time monitoring using sensors and hardware systems.	Mostly hardware-focused; limited dataset analysis and predictive modelling for irrigation planning.	Combine IoT data with analytical models for irrigation decision support.

Key Bibliography

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